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OIL BALANCES WITHOUT HIDDEN ADJUSTMENTS

A Graph-and-Scenario Framework for Crude Oil Logistics Data

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Abstract

Crude oil prices are determined in the short run by the physical balance between supply and demand, with inventories acting as the principal short-run price signal. Producers, refiners, traders, and policymakers therefore spend significant resources reconstructing oil balances from public and commercial data published at multiple resolutions by overlapping agencies. The dominant operational technique — Excel workbooks and Python notebooks that ingest data from several sources, reconcile inconsistencies through hard-coded allocation rules, and balance the result with a residual adjustment column — produces usable numbers but buries inconsistency, hides provenance, and cannot be audited or reused without manual rework.

This thesis proposes a representational framework that treats the physical infrastructure of the oil network as orthogonal to the data attached to it. The physical layer — refineries, pipelines, terminals, basins, storage hubs, vessels — is captured once as a persistent graph of assets. Data binds to that graph through per-scenario assignments: each variable on each asset is either observed (bound to a time series) or derived (computed from a formula over other variables), enforced by a schema-level constraint that makes inconsistent state unrepresentable. Reporting residuals are retained as labelled balancing items on the nodes where they arise, not hidden in adjustment columns. Flows that are physically real but unobserved at the per-edge level are declared as latent variables, distinguished from values that are simply missing.

The framework was implemented on the U.S. crude oil network from January 2015 to December 2024 at monthly resolution, with 251 assets, 433 directed flow edges, and 1,870 variables under the active scenario. A head-to-head comparison against the spreadsheet workflow on the same input data shows that the framework produces a materially different output: the persistent six-million-barrel PADD 5 discrepancy lands on a named in-transit corridor node rather than in an unallocated row; the balancing item at PADD 3 spikes visibly during Hurricane Harvey, COVID-19, the Colonial Pipeline shutdown, and the Strategic Petroleum Reserve release — disruptions that the spreadsheet workflow absorbs silently into the adjustment column. The framework's output is consumable by downstream linear programming and graph neural network models without further cleaning.

Keywords: oil balances · crude oil logistics · graph data models · multi-resolution data · mass balance · schema design · linear programming

Resumo

Os preços do petróleo bruto são determinados no curto prazo pelo equilíbrio físico entre a oferta e a procura, com os inventários a actuarem como o principal sinal de preço de curto prazo. Produtores, refinadores, traders e decisores políticos despendem, por isso, recursos significativos na reconstrução de balanços de petróleo a partir de dados públicos e comerciais publicados em múltiplas resoluções por agências sobrepostas. A técnica operacional dominante — folhas de cálculo em Excel e cadernos em Python que importam dados de várias fontes, reconciliam inconsistências através de regras de afectação codificadas manualmente e equilibram o resultado com uma coluna residual de ajuste — produz números utilizáveis, mas oculta inconsistências, dificulta a auditoria e não pode ser reutilizada sem trabalho manual.

Esta tese propõe um quadro representacional que trata a infra-estrutura física da rede petrolífera como ortogonal aos dados que lhe estão associados. A camada física — refinarias, oleodutos, terminais, bacias, hubs de armazenamento, embarcações — é capturada uma única vez como um grafo persistente de activos. Os dados ligam-se a esse grafo através de afectações por cenário: cada variável de cada activo é observada (ligada a uma série temporal) ou derivada (calculada por uma fórmula sobre outras variáveis), garantido por uma restrição ao nível do esquema que torna estados inconsistentes irrepresentáveis. Os resíduos de reporte são preservados como itens de balanço identificados nos nós onde surgem, e não escondidos em colunas de ajuste. Fluxos fisicamente reais, mas não observados ao nível das arestas, são declarados como variáveis latentes, distintas de valores simplesmente em falta.

O quadro foi implementado para a rede de petróleo bruto dos Estados Unidos entre Janeiro de 2015 e Dezembro de 2024 com resolução mensal, contendo 251 activos, 433 arestas direccionadas de fluxo e 1.870 variáveis sob o cenário activo. Uma comparação directa com o fluxo de trabalho em folha de cálculo, sobre os mesmos dados de entrada, mostra que o quadro produz um resultado materialmente diferente: a discrepância persistente de seis milhões de barris no PADD 5 fica num nó nomeado de corredor em trânsito, em vez de numa linha não afectada; o item de balanço no PADD 3 sobe visivelmente durante o Furacão Harvey, a pandemia de COVID-19, o encerramento do Colonial Pipeline e a libertação da Reserva Estratégica de Petróleo — perturbações que o fluxo de trabalho em folha de cálculo absorve silenciosamente na coluna de ajuste. O output do quadro é consumível por modelos de programação linear e de redes neuronais em grafo sem trabalho adicional de limpeza.

Palavras-chave: balanços de petróleo · logística de petróleo bruto · modelos de dados em grafo · dados multi-resolução · balanço de massa · desenho de esquema · programação linear

Disclosure of Use of Generative Artificial Intelligence

The author, Pedro Porfírio, acknowledges the use of generative artificial intelligence tools in the development of this thesis. These tools were used under direct supervision; all generated outputs were critically reviewed, verified for accuracy, and modified as required. The author accepts full responsibility for the validity and originality of the content, and confirms that the use of these tools is consistent with the institutional policies of NOVA Information Management School and with standards of academic integrity.

Tools used: Anthropic Claude (Sonnet and Opus, multiple versions) and OpenAI ChatGPT (GPT-4 and GPT-5 versions).

Activities supported by these tools: (i) initial enumeration of named physical assets in the U.S. crude oil network (refineries, pipelines, terminals) from publicly available sources, reviewed and reconciled by the author against FERC, PHMSA, OGJ refinery surveys, and trade-press sources; (ii) prose drafting and editing support, where the author retained authorship of all argumentative content; (iii) code-writing assistance for the data-ingest pipelines, the resolver implementation, and the SQL view layer, with the author reviewing every commit; (iv) summarisation support for the literature review, with the author independently verifying each citation against the cited source.

Activities not supported by these tools: the design of the framework's principles, the schema, the resolver's dispatch rules, the consistency claims, the case-study comparison, and the interpretation of all numerical results — these are the work of the author.

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Acronyms

AIS	Automatic Identification System (vessel tracking)
API	American Petroleum Institute (crude grade gravity scale)
ARIMA	AutoRegressive Integrated Moving Average
DCRNN	Diffusion Convolutional Recurrent Neural Network
EIA	U.S. Energy Information Administration
FERC	Federal Energy Regulatory Commission
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GNN	Graph Neural Network
IEA	International Energy Agency
kbbl	Thousand barrels
kbd	Thousand barrels per day
LOCF	Last-Observation-Carried-Forward
LOOP	Louisiana Offshore Oil Port
LP	Linear Programming
mbd	Million barrels per day
mmbbl	Million barrels
MILP	Mixed-Integer Linear Programming
NBER	National Bureau of Economic Research
OGJ	Oil & Gas Journal
PADD	Petroleum Administration for Defense District
PHMSA	Pipeline and Hazardous Materials Safety Administration
SPR	Strategic Petroleum Reserve
STEO	Short-Term Energy Outlook (EIA publication)
STGCN	Spatio-Temporal Graph Convolutional Network
TGAT	Temporal Graph Attention Network
TGN	Temporal Graph Network
VAR	Vector AutoRegression
WPSR	Weekly Petroleum Status Report
WTI	West Texas Intermediate (crude grade benchmark)

Chapter 1

Introduction

1.1 Background and motivation

Crude oil is one of the most volatile commodities in global markets, but it is also one of the most closely monitored. Producers, governments, refiners, and traders interact through spot cargoes, term contracts, and standardised futures linked to regional benchmarks. Even so, those financial instruments ultimately depend on the physical balance between supply and demand for crude itself. For that reason, serious quantitative work on oil markets usually starts with the physical system that produces, transports, stores, and consumes the commodity.

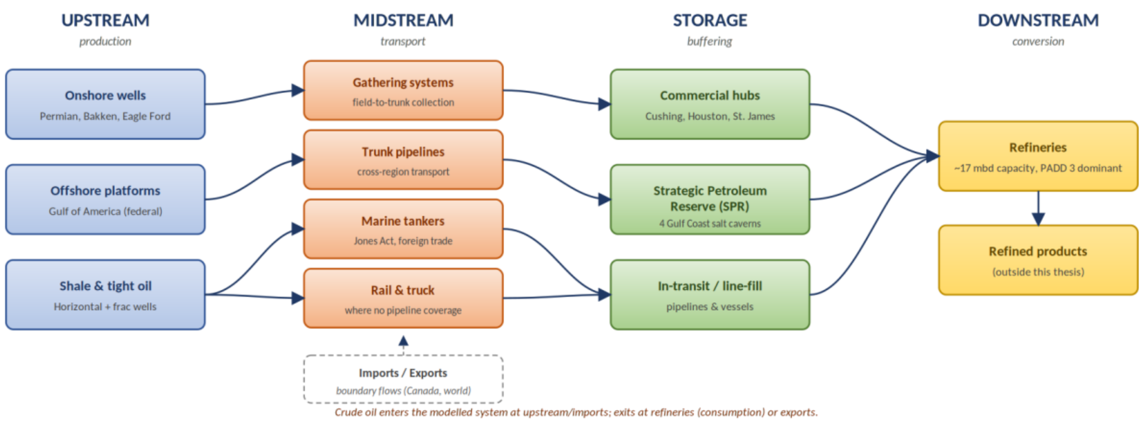

Figure 1.1. *The crude oil physical supply chain: upstream production (onshore wells, offshore platforms, shale and tight oil), through midstream gathering and transport (pipelines, tankers, rail and truck), through storage (terminals and SPR), into downstream refining and refined products.*

In the short run, the physical layer of an oil network is rather constant. On the upstream side, well output is constrained by geology, reservoir conditions, and capital that was committed years before the current period. The economic literature is explicit on the consequence: the short-run price elasticity of oil supply is very low, because there is a lag between investment and production, the costs of adjusting extraction rates are convex, and existing fields show decreasing returns (Bornstein, Krusell, and Rebelo, 2017; Hamilton, 2009). On the downstream side, refineries operate in narrow design windows. They are built for particular throughput ranges and crude slates, and short-term operation is structured around a small set of discrete modes with significant transition cost; this is why operations-research treatments of refineries model mode selection rather than continuous throughput control (Pinto, Joly, and Moro, 2000; Joly, Moro, and Pinto, 2002; Fernandes, Relvas, and Barbosa-Póvoa, 2013). Industry data is consistent with this picture: U.S. refinery utilization rates have hovered between 87% and 95% for most of the past decade, with a practical lower limit — the “turndown rate” — of approximately 65–70% below which units cannot run economically. Transport adds further delay, with pipelines, ports, and tankers introducing transit times of days to months (Stopford, 2009; Kazemi and Szmerekovsky, 2015). Production and consumption therefore hold close to their operating set-points most of the time, and meaningful changes on either side usually require lead time, capital, and coordination across several parts of the chain (Lima, Relvas, and Barbosa-Póvoa, 2016).

Most short-term mismatches between supply and demand are step changes — shocks to one side that the other side cannot follow. A refinery enters unplanned maintenance, and its crude intake drops abruptly; the upstream production feeding it does not slow correspondingly, because wells and gathering systems cannot be throttled on a few days' notice. A hurricane shuts platforms and refineries on the Gulf Coast at the same time, but the imbalance between the two sides of the balance is rarely symmetric. A pipeline outage interrupts a particular corridor while production at its upstream end and demand at its downstream end carry on as before. In each case, one side of the balance step-changes while the other side stays put, because that is what physical infrastructure does in the short run.

Inventories absorb the resulting gap, and the change in stocks is the observable signature of every such shock. Tank farms, storage hubs, and strategic petroleum reserves are the buffer between what is produced and what is consumed at any given moment, and the level of stocks is therefore one of the principal determinants of short-term crude price dynamics (Working, 1949; Pindyck, 2001; Kilian and Murphy, 2014; U.S. Energy Information Administration, 2024). When stocks are comfortable, shocks can be absorbed with limited price impact. When they become tight, prices rise; when they become excessive, prices soften; and when usable storage is effectively full, the marginal barrel has nowhere to go. The April 2020 WTI front-month settlement at −37 USD per barrel — a price at which the seller pays the buyer to take delivery — was a highly visible demonstration of that last limit. Because the mismatches arrive as shocks rather than as drift, inventory changes are not noise around a trend; they are the recoverable trace of identifiable events, and reading them correctly is what makes a balance informative.

Two ideas from Pindyck (2001) make this connection precise and are particularly load-bearing for the work that follows. The first is that inventories act as the buffer that decouples production from consumption, so spot prices reflect the flow imbalance between the two rather than the level of either; price formation is therefore inseparable from the dynamics of stocks. The second is that production facilities — wells, refineries, and the pipelines that connect them — behave as call options on the underlying commodity, which is why their economic value responds to volatility the way it does. Both ideas map directly onto the framework developed in this thesis: inventories appear as explicit state variables at every physical node, and the structural identity of each producing or consuming asset is preserved as a first-class entity rather than collapsed into aggregate flow.

The operational consequence is direct: if one cannot measure how inventories change, one cannot understand what is happening to price. Stock changes reveal whether a shock has been absorbed by drawing down storage or has spilled into the market, whether refinery throughput is keeping pace with crude availability, and whether a flow that nominally took place has arrived at its destination. Producing oil balances that track stock changes correctly is therefore not a back-office accounting exercise; it is the upstream condition for any quantitative work on oil prices.

That upstream condition is harder to meet than it appears, because the data does not arrive in a shape that matches the physical flow. Public and commercial sources publish at granularities that do not line up with the operational network. The U.S. Energy Information Administration releases monthly series at national, district (PADD), refining-district, state, and named-basin levels. The International Energy Agency publishes country and regional aggregates. Commercial providers — Genscape, Kpler, IIR Energy, S&P Global Commodity Insights — add higher-frequency and finer-granularity views from satellite imagery, AIS feeds, and operator surveys. The same physical quantity often appears at several resolutions at once, and the resolutions do not always reconcile cleanly. Reporting boundaries are administrative rather than operational; some flows are directly measured, and others can only be inferred from aggregate constraints; inventories often close only at coarser levels than the production they are supposed to balance against; and the topology itself changes over time through real events such as pipeline reversals, terminal commissioning, and refinery shutdowns. The structural mismatch between how the system flows and how the data describes it is the central friction this thesis addresses.

The current state of the art for handling that friction in practice is ad hoc. Analysts maintain Excel workbooks or pandas notebooks in which production, consumption, flows, and stocks are loaded from multiple sources, reconciled through hard-coded allocation rules, and balanced with a residual term that absorbs whatever does not add up. Those workflows produce usable numbers, but they bury inconsistency inside the residual and embed each analyst's assumptions in formula cells that cannot be audited from the outside. They cannot be passed directly into an optimisation model or a machine-learning pipeline without further hand-cleaning, and two analysts working from the same data routinely produce different outputs.

The balancing item. The residual term that appears at the end of every oil balance deserves explicit attention from the outset, because it is both unavoidable and informative. The International Energy Agency already publishes a balancing item in its supply–demand tables, defined as the gap between observed stock changes and the sum of observed flows. The convention treats that gap as data: a labelled variable on the published balance, whose size and sign reveal where the reporting system is and is not closing cleanly. When the practitioner spreadsheet workflow hides the same residual in an unlabelled adjustment column, that information is lost — the bottom row of the balance reads zero by construction, but the signal that produced the residual is no longer recoverable. A representation that aims to support quantitative work on inventories must therefore not only track production, consumption, and flows; it must also retain the balancing item as a first-class variable wherever the data fails to close. Predicting changes in inventories means, equivalently, predicting the production and consumption flows that drive them and the balancing item that accommodates whatever the data cannot resolve.

Modelling tools used to study systems like this fall into three broad groups. Operations research — linear, mixed-integer, and stochastic programming — is mathematically rigorous and well suited to strategic design and tactical planning, but it usually relies on simplified, bespoke representations that leave out much of the operational detail visible in the real network. Classical time-series methods such as ARIMA, VAR, and more recent neural forecasting models are typically applied to individual variables — refinery throughput, regional inventories, benchmark prices — without explicitly modelling the network structure that links those variables together. Over the last decade a third line of work has emerged at their intersection. Graph neural networks and graph representation learning offer a way to model systems whose behaviour depends both on local time dynamics and on a broader relational structure, and recent surveys (Wu et al., 2020; Jin et al., 2024) document how quickly this literature has expanded across traffic networks, power systems, and, more recently, supply chains (Kosasih and Brintrup, 2022; Wasi et al., 2024; Ahn et al., 2024). What makes these methods especially relevant here is that they can bring together two aspects of the problem that classical operations research and standard time-series approaches often handle separately: the network itself and the temporal behaviour of the variables attached to it.

Before any of these modelling tools can be applied to the inventory problem, however, the same upstream question appears: how should the network itself be represented? A representation that can host multi-resolution data without forced reconciliation, that retains the balancing item as a labelled variable rather than absorbing it into an adjustment column, and that admits structural change without requiring a wholesale rebuild, would be useful not only for the optimisation and forecasting work positioned as the framework's downstream consumers, but also for the analysts who currently connect public data to those models through manual workflows.

1.2 Problem statement

Despite the maturity of the three research streams outlined above, there is still no widely adopted way of representing crude oil logistics networks that meets the needs of downstream users. Existing approaches do not deal cleanly with multi-resolution data without relying on ad hoc reconciliation. They do not enforce single attribution at schema level, do not preserve the provenance of derived quantities across the full resolution chain, and do not clearly separate the slowly changing physical topology from the faster-changing data assignments attached to it.

Anyone who has worked with energy data will recognise the practitioner workflow that fills this gap. Analysts often maintain Excel workbooks or pandas notebooks where production, consumption, flows, and stocks for each PADD are loaded from a multitude of sources, reconciled through hard-coded allocation rules, and balanced with a residual adjustment term that absorbs whatever does not add up. Those workflows can produce useful numbers, and they are deeply embedded in day-to-day market analysis, but they do so by burying inconsistency inside the adjustment term. They are also difficult to generalise: each workbook reflects its author's assumptions, and none can be passed directly into an optimisation model or a machine-learning pipeline without further manual cleaning.

The central problem this thesis addresses is therefore: can we design a graph representation of the physical flow of oil that is permanent in structure, accepts different combinations of multi-resolution data, and supports the prediction of changes in inventories — including the balancing item that absorbs whatever the data cannot resolve — in a form consumable by classical optimisation models (linear programmes) and by modern machine-learning models (graph neural networks) without further hand-cleaning?

This thesis answers that question by developing, implementing, and validating an asset-centric temporal graph framework organised around a small set of design principles, together with a deterministic resolver that produces a single per-(scenario, variable, observation date) value table that downstream consumers can read directly. The framework is then demonstrated head-to-head against the dominant current technique on the same input data.

1.3 Research objectives

The research pursues three main objectives:

To design a graph schema that represents the crude oil logistics network at the level of physical assets, captures both structural relationships and time-series dynamics, enforces single attribution and partition closure as schema-level guarantees, and accommodates multi-resolution public data without treating reconciliation residuals as primary data.

To implement that schema as a working system, including a relational database representation, a deterministic resolver that produces the per-(variable, date) data tensor, a layered view structure for downstream consumers, and a complete starter instance covering the U.S. crude oil network from 2015 onwards.

To demonstrate that the framework produces materially more honest balances than the dominant current technique on the same input data, through a head-to-head case study in steady state and under stress, and to assess its readiness for further downstream consumers such as graph neural networks for forecasting, which are left for future work.

1.4 Scope and delimitations

This research focuses on the representational framework and its first downstream consumers. The asset graph is built at the level of named physical assets wherever public data allows, with residuals used for un-named tail production. The geographic scope is the United States, the temporal resolution is monthly, and the initial implementation treats crude oil as a single commodity grade. The schema is designed to be extendable along all three dimensions — to other grades and refined products, to other countries, and to finer temporal granularity — and those extension paths are discussed where relevant. Even so, the implemented instance in this thesis is the U.S. monthly crude oil graph from January 2015 to December 2024 inclusive.

Two important areas are explicitly outside the scope of the present work and are positioned as future work in Chapter 6. The first is forecasting. Forecasting is treated in this thesis as a downstream use case that the framework is designed to support but does not itself implement. Chapter 6 explains why the framework is ready for forecasting work and outlines what that next step would involve. The second area is production deployment. The implementation developed here is a working research artefact rather than a production system, so issues such as hardware and software selection, user interfaces, monitoring, and access control are left aside.

Two pieces of vocabulary used throughout deserve flagging upfront. The framework distinguishes assets from nodes. An asset is a primitive entity: a refinery, a pipeline, a basin, an export terminal. A node is how that asset appears in a particular graph. Each node binds to exactly one asset, and the same asset can appear in multiple graphs without being duplicated in the underlying data. The starter scenario uses a single graph, so in the current implementation each asset has exactly one node, but the separation is designed in: it lets per-grade, per-operator, or per-period views over the same physical reality be added later without restructuring the asset registry.

Assets carry a kind flag with two values. A physical asset is a real, named entity with capacity, geographic coordinates, configuration, and flow edges to its neighbours; refineries, pipelines, terminals, basins, gathering networks, and storage hubs are all physical assets. An abstract asset is an aggregation view over the physical layer (observational aggregates such as district views, basin views, refining-district views, and stock-decomposition aggregates) and has no flow edges of its own. Abstract assets exist to host roll-up observations and partition relationships and constraints of other nodes; their relationship to the physical layer is declared through formula inputs rather than through edges. Mass balance is enforced at every physical asset, and abstract assets inherit it as a corollary of aggregation — they do not introduce intrinsic constraints unless explicitly stated.

Two extensions are worth flagging here because they recur in conversations about the framework. The first is per-grade decomposition: by treating commodity as a further dimension on every variable, each tank farm carries inventory per (asset, grade), and the evolving blend in storage is recoverable as the running sum of grade-specific inflows minus grade-specific offtakes — a per-(tank, grade) inventory series rather than a single aggregate. The second is refinery yield: the framework expresses refined-product output as a derived variable through formula and formula inputs, so a refinery's gasoline production can be defined as a fixed fraction of its consumption of a particular grade (gasoline production = 0.35 × crude consumption of grade X), with the yield coefficients themselves potentially scenario-specific. Both extensions sit on top of the schema as constructed; neither requires structural change.

1.5 Thesis structure

The remainder of the thesis is organised as follows. Chapter 2 reviews the techniques currently used to produce oil balances — official statistical balances, the practitioner spreadsheet workflow, optimisation models, and graph and machine-learning approaches — and identifies the gap that the thesis addresses. Chapter 3 gives a focused domain background on the U.S. crude oil network, with particular attention to its multi-resolution reporting structure.

Chapter 4 presents the framework as five design principles, organised around the orthogonality of physical infrastructure and data. Chapter 5 is the central comparative chapter: a head-to-head case study against the spreadsheet workflow on the same input data, in steady state and under stress. Chapter 6 concludes with the contribution, the limitations, and a discussion of future work, including the framework's readiness for graph neural network forecasting as a natural next consumer.

Three annexes provide supporting reference material. Annex A documents the implementation — the schema, the starter U.S. crude oil graph, the resolver, and the consistency audits. Annex B catalogues the variables included in the starter scenario. Annex C offers a primer on graph neural networks for readers who want to follow the GNN-readiness discussion in Chapter 6 in more detail.

Chapter 2

Current techniques for producing oil balances

This chapter reviews the techniques in current use for producing oil balances. Each technique is described on its own terms, with its strengths acknowledged, and its limitations stated. The chapter is organised around four families: official statistical balances published by national and international agencies; the practitioner workflow built on spreadsheets and notebooks; optimisation models from the operations research literature; and graph and machine learning approaches from the recent computer science literature. The chapter closes with the gap that the thesis addresses.

2.1 Official statistical balances

The U.S. Energy Information Administration is the dominant public source for the U.S. crude oil network. Its Monthly Petroleum Supply Reporting System publishes production, refinery runs, stocks, imports, exports, and inter-district movements at several resolutions: national totals, the five Petroleum Administration for Defence Districts (PADDs), refining districts that subdivide each PADD, and named basins and states within each PADD. The Weekly Petroleum Status Report provides a higher-frequency view of a smaller set of series. Across this catalogue, the same physical quantity is often published at more than one resolution simultaneously, with reconciliation between the levels left to the user.

The International Energy Agency follows a similar pattern at the international level. Its monthly Oil Market Report and annual World Energy Balances present supply, demand, refining, stocks, and trade for OECD countries and major non-OECD producers. A point of methodology that matters for what follows: the IEA retains an explicit balancing item in its supply–demand tables, defined as the residual between observed stock changes and the sum of observed flows. The convention treats the residual as information rather than as noise to be hidden. The size and sign of the balancing item indicate where the reporting system is not closing cleanly, and the IEA does not try to make that issue disappear by silently folding it into nearby quantities. This thesis follows the same convention, applied at every physical node in the network rather than only at country-level aggregates.

Official statistical balances have well-known strengths: standardised methodology, free access, international comparability, decades of historical coverage. Their limitations are also well known. The monthly granularity is too coarse for several operational questions; reporting boundaries are fixed by administrative convenience rather than by physical structure; named facilities are exposed only patchily, with most series aggregated to the PADD or refining-district level; and the same publication often contains series at incompatible resolutions, with no canonical guidance on which is authoritative when they disagree. The published balance is a useful starting point for any further analysis, but it is not by itself the operational answer that traders, refiners, or analysts need.

2.2 The practitioner spreadsheet workflow

The dominant operational technique for producing oil balances inside trading firms, consultancies, integrated oil companies, and government agencies is a workbook or notebook that ingests the relevant published series, reconciles them across resolutions through a set of allocation rules embedded in formulas, and presents a balance to the user. Anyone who has worked with energy data will recognise the pattern. Production at named basins is summed into a PADD total, which is compared against the published PADD aggregate. Imports and exports per PADD are read from the relevant series and combined with refinery throughput and stock changes. Whatever does not add up is absorbed into a residual column, usually labelled “adjustment”, “statistical difference”, “unallocated”, or simply left without a label. The bottom row of the balance reads zero by construction.

These workflows have real strengths. They are fast to build, easy to modify, embedded in the daily work of the people who use them, and produce a usable answer in seconds. They also accumulate institutional knowledge: a workbook that has been refined over years of trading desk operation often encodes subtleties about EIA's revision conventions, custody-transfer timing, or particular pipelines that are not written down anywhere else. The technique is not a methodological mistake; it is the natural response to the underlying data problem, and the framework developed in this thesis is best understood as a more disciplined version of the same instinct.

Their limitations are equally real. The first is that reconciliation rules are implicit and inconsistent across analysts. Each workbook embeds its author's assumptions about how to allocate the PADD-level production residual across named basins, how to handle the difference between gross and net imports, how to attribute the SPR drawdown to commercial stocks. Two analysts working on the same EIA data produce different derived quantities because their reconciliation choices differ, and the differences are not visible in the bottom line.

The second limitation is that the residual adjustment column hides information. The IEA's convention of publishing the balancing item at the national level exists because reporting frictions are systemic — custody-transfer timing, refinery process losses, vessel measurement differences, in-transit allocations — and the size of the residual carries information about where the reporting system is failing to close. A workbook that absorbs the residual into an unlabelled adjustment cell loses that signal. The bottom row reads zero, but the analyst no longer knows whether the period balanced because the data is clean or because the adjustment column did its work.

The third limitation is provenance. A derived quantity in a typical workbook depends on a chain of allocations and adjustments distributed across dozens of cells, and tracing any specific value back to its inputs requires reverse engineering the workbook. This makes the workflow hard to audit, hard to hand over to a new analyst, and hard to extend to new data sources. When EIA introduces a finer-resolution series — for example, when PADD-level stocks were decomposed into commercial and SPR portions — incorporating it into a legacy workbook requires modifying the allocation logic in several places, often introducing inconsistencies with the resolution that was previously authoritative.

The case study in Chapter 5 compares the framework developed in this thesis against this workflow directly, on the same input data, in two specific scenarios. The comparison makes the three limitations concrete: a structural discrepancy that the workflow leaves in an unlabelled adjustment cell lands on a named node in the framework, and a disruption month that the workflow absorbs silently into the adjustment column produces a visible spike in a labelled balancing item in the framework.

2.3 Optimisation models

A second family of techniques approaches the network through mathematical programming. The literature on crude oil and refined-product supply chains is extensive, dominated by mixed-integer linear programming and its stochastic extensions. Fernandes, Relvas, and Barbosa-Póvoa (2013) develop a deterministic MILP for the strategic design of multi-entity, multi-echelon, multi-product downstream petroleum supply chains. Kazemi and Szmerekovsky (2015) extend this line of work to multimodal transportation, showing that the joint treatment of pipeline, rail, marine, and truck transport can materially change the optimal solution compared with single-mode formulations. Ghaithan, Attia, and Duffuaa (2017) move toward tactical planning with a multi-objective formulation in an integrated oil and gas setting. On the uncertainty side, Ribas, Hamacher, and Street (2010) develop stochastic programming models for integrated oil-chain planning under uncertainty. Lima, Relvas, and Barbosa-Póvoa (2016) provide a broad review of the downstream oil supply chain management literature, noting that the level of detail at which optimisation models are built often does not match the level of detail at which real input data is available.

The methodological strength of this literature is that the formulations are mathematically rigorous and supported by mature solver infrastructure. The structural commitment that runs through the family is the level of abstraction at which the network is represented: nodes are typically large aggregates (refineries, demand centres, production regions), and edges are abstract connections defined in terms of cost, capacity, or lead time. That representation is well suited to the strategic and tactical questions these models are designed to answer, such as where to locate capacity or how to allocate supply across regions. It is less helpful for the upstream question this thesis addresses, which is how to represent the network in a way that admits multi-resolution data and supports several different downstream consumers without modification. Each optimisation paper builds its own bespoke representation; the representation is a means to the model's end, not a research artefact in its own right.

2.4 Graph and machine learning approaches

A third family of techniques applies graph data models and graph representation learning to commodity and logistics networks. The applied literature is recent. Kosasih and Brintrup (2022) show that graph neural networks can predict hidden links in supply chain networks, revealing dependencies that aggregate-level data tends to miss. Wasi, Islam, and others (2024) introduce SupplyGraph, the first benchmark dataset explicitly designed for GNN-based supply chain analytics, with tasks including demand forecasting, production planning, and product classification. Ahn, Yoon, and Park (2024) develop a probabilistic GNN model for supply and inventory prediction using data from a global consumer-goods company, showing that attention-based graph architectures can improve predictive performance by capturing cascading effects across nodes.

Behind these applied studies sits a methodological literature on graph representation learning. Foundational work on random-walk embeddings (Perozzi, Al-Rfou, and Skiena, 2014; Grover and Leskovec, 2016) and on graph neural networks more directly (Kipf and Welling, 2017; Veličković et al., 2018; Hamilton, Ying, and Leskovec, 2017) established the methodological base. Surveys by Wu et al. (2020) and Jin et al. (2024) provide useful overviews of the field. Temporal extensions such as TGAT (Xu et al., 2020), TGN (Rossi et al., 2020), and hybrid architectures like DCRNN (Li et al., 2018) and STGCN (Yu, Yin, and Zhu, 2018) make the temporal dimension an explicit part of the modelling framework. The traffic-forecasting literature in particular has produced architectures that fit logistics networks structurally well, because traffic networks share with crude oil networks a largely stable topology, clearly identified nodes, and forecasting targets that depend both on local time dynamics and on dependencies that propagate through the network.

These studies are important because they demonstrate that graph methods can add value in commodity and logistics settings beyond what isolated node-level time-series models provide. They share, however, a limitation that matters for the question this thesis asks. Each is built around its own application-specific data representation; the representation is tailored to the task, not treated as a contribution. SupplyGraph is structured around production lines and demand centres in fast-moving consumer-goods setting; its node types, edge meanings, and temporal resolution make sense there but do not transfer directly to crude oil. The same point applies to the other applied papers. The methodological literature is largely silent on the upstream question of how to construct the data representation in the first place, taking it as given.

2.5 The gap

The three streams reviewed above are each useful for the questions they address, and none addresses the question this thesis poses. Official statistical balances publish monthly aggregates at fixed resolutions, with no canonical way to reconcile them across levels. The practitioner spreadsheet workflow produces operational balances quickly but buries inconsistency in unlabelled adjustment columns. Optimisation models work at a level of abstraction that leaves out the multi-resolution structure of the data. Graph and machine learning approaches take the upstream representation as given and tailor it to the task.

What is missing across all four is a representation that combines four properties at once: it admits multi-resolution data without ad-hoc reconciliation; it enforces single attribution at the schema level rather than through post-hoc checks; it preserves the provenance of derived quantities so that any value can be traced back to its sources; and it produces output that several different downstream consumers can read without modification. The thesis fills that gap by developing such a representation, implementing it on the U.S. crude oil network, and demonstrating that it produces materially more honest balances than the dominant current technique on the same input data.

Chapter 3

Domain background

This chapter provides the domain background needed to make the design choices in Chapters 4 and 5 comprehensible. It is deliberately focused: just enough description of the U.S. crude oil system to establish vocabulary, with the most attention given to the multi-resolution reporting structure that the framework responds to. Readers who want a full industry primer are referred to Stopford (2009) for the maritime side and to the EIA's published methodology documents for the statistical side.

3.1 The U.S. crude oil network

At the scale of the United States, the crude oil system can be described in terms of four physical layers: upstream production, midstream transport, storage, and downstream refining. The upstream layer brings crude to the surface. Onshore conventional production, offshore production concentrated in the federal waters of the Gulf of America (formerly the Gulf of Mexico), and shale and tight oil production drive U.S. output. The Permian Basin in Texas and New Mexico, the Bakken in North Dakota and Montana, and the Eagle Ford in Texas are the largest producing regions; together with offshore Gulf production, they account for the majority of U.S. crude.

The midstream layer moves crude from production regions to storage, refining centres, or export terminals. Gathering systems collect output from individual wells; trunk pipelines move crude over longer distances; marine transport handles international movements and, through the Jones Act fleet, certain domestic flows between PADDs; rail and truck handle smaller volumes where pipeline coverage is limited. The storage layer absorbs short-term imbalance. Commercial tank farms at major hubs — Cushing in Oklahoma, Patoka in Illinois, Houston, St. James, the LOOP system offshore Louisiana — provide the working storage used by refiners and traders. Alongside them sits the Strategic Petroleum Reserve, a set of four underground salt-cavern sites on the Gulf Coast operated by the federal government for emergency supply.

The downstream layer is where crude is converted into refined products. U.S. refining capacity is approximately 18 million barrels per day, with actual crude runs typically in the 15–17 million range. Capacity is heavily concentrated in PADD 3 on the Gulf Coast, which accounts for roughly half the U.S. total. For the purposes of this thesis the downstream layer marks the point at which crude exits the modelled system and reappears as refined products; the refining yield problem is in scope as a future extension but is not implemented in the active scenario.

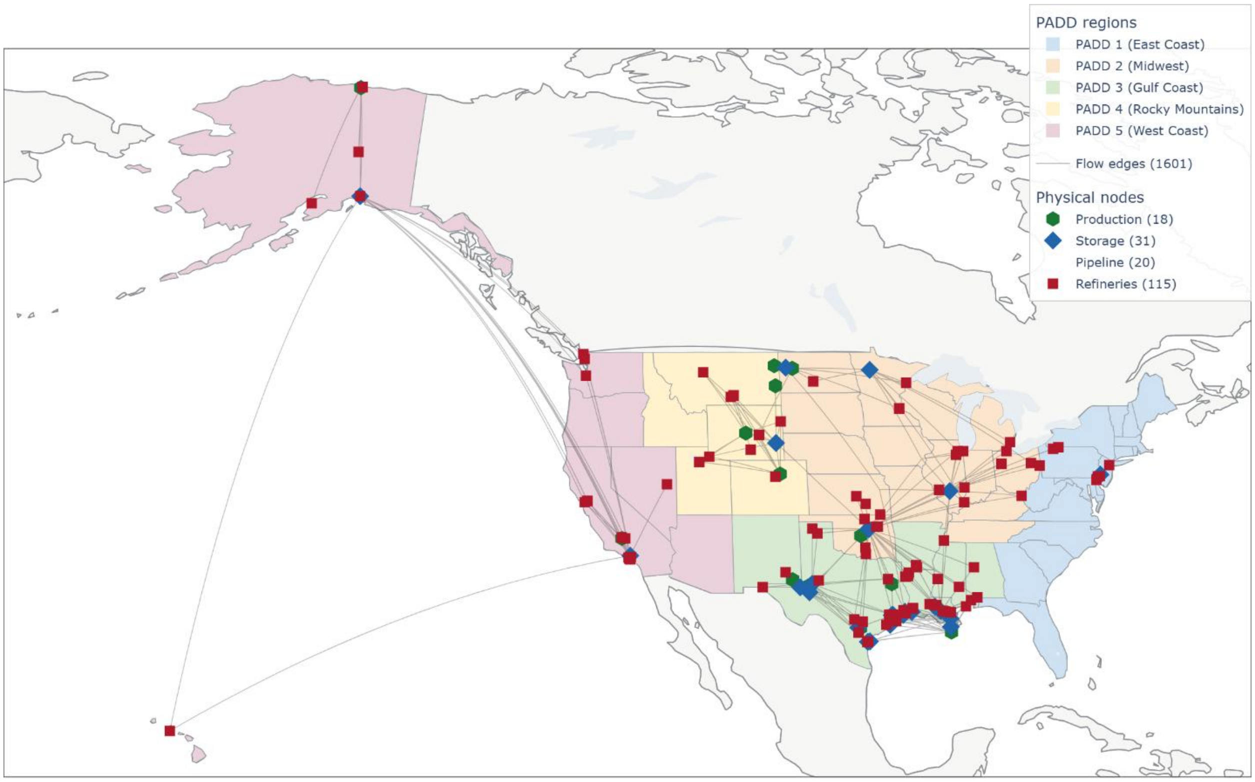

Figure 3.1. *U.S. crude oil transport network as instantiated in the framework. Production basins (green) feed gathering systems and origin terminals. Storage hubs (dark blue) sit at the major junctions; SPR sites (yellow) cluster on the Gulf coast. Refineries are shown in red, import and export terminals in orange. Latent per-edge flows from Permian to its three origin terminals are shown as grey dashed arrows. The PADD-3-to-PADD-5 in-transit corridor (Section 5.2) is highlighted in red.*

3.2 The multi-resolution reporting structure

The U.S. Energy Information Administration organises petroleum statistics around five Petroleum Administration for Defense Districts (PADDs). The districts were created in 1942 for wartime fuel rationing and have remained the dominant geographic structure for U.S. petroleum reporting ever since. PADD 1 covers the East Coast; PADD 2 the Midwest; PADD 3 the Gulf Coast; PADD 4 the Rocky Mountain region; PADD 5 the West Coast. Within each PADD, EIA reports refinery activity at the level of refining districts (PADD 3A, 3B, 3C, and so on) and production at the level of named basins and states.

The multi-resolution structure of EIA's publications is the central feature of the data that the framework must accommodate. The same physical quantity often appears at several resolutions at once, and the resolutions do not always agree. Table 3.1 summarises how the principal balance components are published.

Component	Resolution(s) published	Notes
Production	USA total, PADD, refining district, state, named basin	Same physical output reported at every level; named-basin sum within a PADD can differ from the PADD total by several hundred kbd.
Refinery runs	USA total, PADD, refining district	Facility-level data only for PADD 3 sub-districts.
Stocks	USA total, PADD, named hub (selected), SPR sites	Commercial, SPR, and in-transit reported separately at PADD level.
Imports	USA total, PADD, by country of origin	Per-PADD-per-country granularity; no per-terminal allocation.
Exports	USA total, PADD, by country of destination	Per-PADD-per-country granularity; no per-terminal allocation.
Inter-PADD flows	PADD pair (e.g. PADD 2 → PADD 3)	No per-pipeline allocation; many physical pipelines aggregated.

Table 3.1. *Principal balance components published by EIA at multiple resolutions. The same physical quantity is reported at several levels at once, with no canonical way to reconcile across levels.*

Three features of this structure matter for what follows. The first is that production at a named basin within a PADD does not in general add up to the published PADD total, because EIA includes some production in the PADD aggregate that is not separately named at the finer level. The second is that flows are reported at the level of PADD pairs rather than per pipeline, so the individual contribution of, say, the Capline or Diamond pipeline to PADD-2-to-PADD-3 movement is not publicly observed. The third is that several flows are physically real but not directly observed at any resolution — the per-terminal split of Permian outflows across Midland, Wink, and Crane is the canonical example. These three features are direct constraints on the representation.

3.3 What a representation must accommodate

The physical and reporting structure described above yields five requirements that the framework must meet:

- **Multi-resolution data without forced disaggregation.** Production observed at basin level, stocks at PADD level, consumption at refining-district level, and inter-PADD flows at PADD-pair level must coexist in a single representation. The framework must not require any of these to be "down-disaggregated" to a uniform finest-granularity view before being usable.
- **Junction fan-outs with unobserved per-edge splits.** The Permian outflow problem (one production node, three origin terminals, multiple pipelines from each, downstream destinations

observed only at aggregate level) is structural, not exceptional. Bakken gathering, the Cushing hub, and the Gulf Coast export complex all exhibit the same pattern at different scales.

- **Residuals as information, not noise.** The IEA convention of publishing the balancing item exists because the data does not close exactly, and the residual carries information about where the reporting system is failing to close. The framework must retain the residual as a first-class variable rather than absorbing it into adjacent flows.
- **Slow structural evolution.** The pipeline network changes deliberately over time — the Capline pipeline reversed direction in 2022, and a Bakken cross-state midstream pool came into operation in 2024. These are real operational events. The framework must accommodate them without requiring a wholesale graph rebuild every time data resolution improves.
- **A clear system boundary.** Foreign supply enters through Canadian and other import flows; U.S. crude leaves through export terminals to destinations that the framework does not track. The boundary needs to be explicit, with mass balance closing on the U.S. side without requiring the framework to model the foreign side.

These five requirements are not optional additions. They follow directly from the way the system operates and the way the data is published. The next chapter introduces the framework that responds to them.

Chapter 4

A graph-and-scenario framework

This chapter presents the representational framework as five design principles. Each principle is stated, explained in terms of what it enables, and shown to respond directly to one or more of the requirements identified at the end of Chapter 3. Together the five principles produce a representation in which every value traces to one source, aggregates close exactly to their constituents, mass balance holds at every physical node, and reporting residuals are labelled rather than hidden.

4.1 Physical infrastructure is orthogonal to data

> **Principle 1 — Orthogonality.** Keep the physical layer and the data layer separate. Data updates must not change the graph; structural changes must not require data revisions.

The framework's central commitment is a separation between two layers that current techniques conflate.

The physical layer of the oil network changes slowly. New refineries open every few years; pipelines reverse direction every few decades (Capline reversed in 2022, Seaway in 2012); terminals are commissioned and retired on multi-year timescales. Each structural change reflects a real operational event in the physical world, often involving substantial capital expenditure, and is rare on the scale of a working scenario.

The data layer changes constantly, and at different rates for different variables. EIA publishes monthly series at the start of each month. Commercial providers update vessel positions every few minutes through AIS. Tank-level inventory at named hubs may be updated daily from satellite imagery; refinery operating status may be revised weekly. The granularity at which any given quantity is observed also varies: a flow that is reported today only at PADD-pair level may be reported tomorrow at per-pipeline level when a new commercial source becomes available.

Treating these two layers as a single thing forces every data improvement to rewrite the graph and every structural change to revisit the data. A workbook that incorporates a new finer-resolution series needs to modify the allocation logic that previously distributed the coarser figure; a graph that hard-codes the topology to match a particular dataset becomes obsolete when the dataset changes. The orthogonality principle resolves this by keeping the two layers separate. The physical graph is built once from public infrastructure data — corporate filings, FERC and PHMSA registers, OGJ refinery surveys, trade-press surveys — and updated only when a real physical event takes place. Data binds to that graph through per-scenario assignments, which evolve continuously as new series and new sources become available, without ever modifying the underlying topology.

This orthogonality is the framework's primary contribution. The four principles that follow are operational consequences: each is necessary for the orthogonality to be workable in practice.

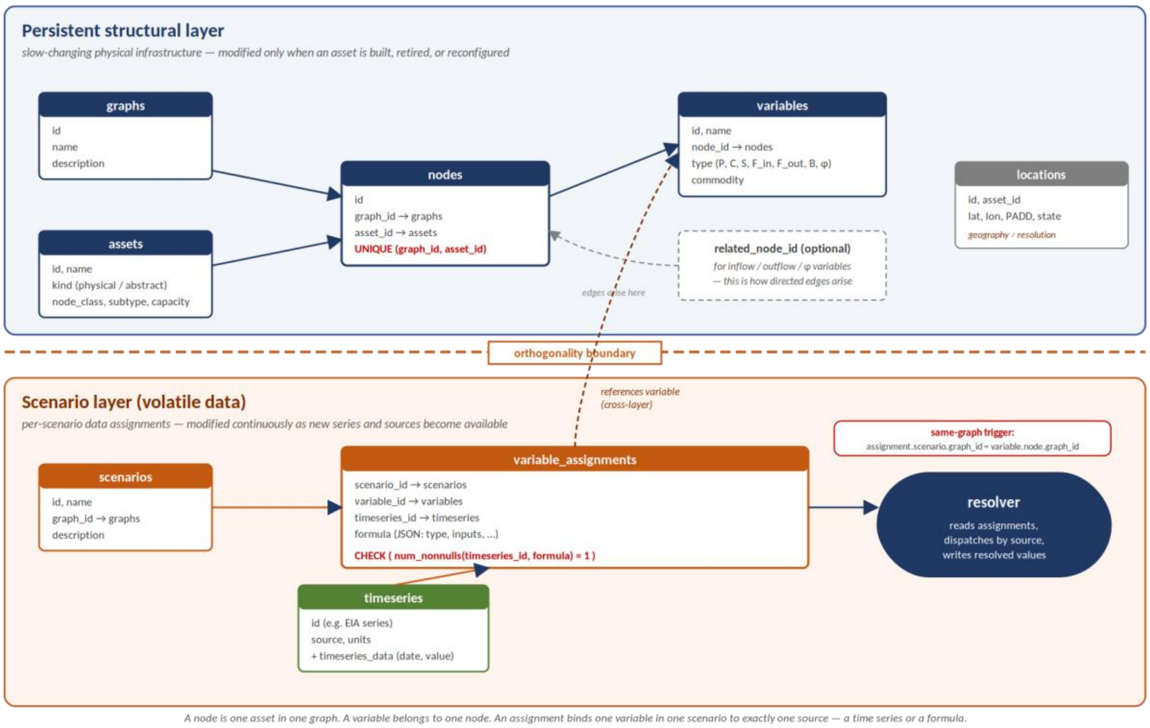

Figure 4.1. *The oil_network data model as enforced by the schema. The persistent structural layer (top, blue) holds graphs, assets, nodes, variables, and locations — the slow-changing physical infrastructure. The scenario layer (bottom, orange) holds scenarios, variable_assignments, and timeseries — the volatile data attached to that infrastructure. The CHECK constraint num_nonnulls(timeseries_id, formula) = 1 on variable_assignments enforces single attribution at the schema level. The same-graph trigger ensures that an assignment's scenario and its variable's node resolve to the same graph. The resolver consumes assignments and produces resolved values for downstream consumers.*

4.2 Assets are first-class nodes

Principle 2 — Assets as nodes. Every physical asset — refinery, pipeline, vessel, terminal, basin, gathering system — is a node carrying its own inventory and mass balance. Edges are flows; assets are not edge attributes.

Every physical asset is represented as a node in the graph, with its own inventory variable and its own mass balance. Refineries, pipelines, vessels, terminals, basins, gathering systems, and storage hubs are all first-class nodes. Edges represent the directed movement of crude from one asset to another; they carry volume, but they do not hold it.

The alternative — a location-centric representation in which nodes denote places and assets sit on the edges between them — is common in the optimisation literature and well suited to strategic questions about where to locate capacity. It is unsuited to the question of where barrels actually are at any given moment. A trunk pipeline holds several days of throughput as line-fill, an amount comparable to a regional refinery's weekly runs; that line-fill is hidden state in a location-centric model, reconstructed indirectly from transit time and recent flows. A very large crude carrier holds two million barrels for the duration of its voyage; treating that volume as an edge attribute makes the cargo invisible to any analysis that asks where U.S. stocks actually sit. The asset-centric commitment exposes both: the pipeline's line-fill is its own stock variable, the vessel's cargo is its own stock variable, each accountable as the system makes them accountable in operational practice.

The same framing pays off when richer data sources are introduced. Genscape, now part of Wood Mackenzie, reports daily tank-by-tank inventory at the Cushing hub from infrared satellite imagery — roughly twenty individual storage tanks resolved rather than a single PADD-2 aggregate. Kpler tracks vessel cargoes globally through AIS feeds, resolving in-transit crude by ship, route, and grade. IIR Energy maintains a facility-level refinery operations database. In the asset-centric representation, each of these sources binds to the node it describes; in a location-centric representation, where pipelines and tanks are edge attributes, none of them would have a natural home.

4.3 Mass balance with an explicit balancing item

> **Principle 3 — Mass balance with B.** Every node satisfies $\Delta S = P + \Sigma F_{in} - C - \Sigma F_{out} + B$. The balancing item B is a labelled, first-class variable — never an unlabelled adjustment column.

Every physical node in the framework satisfies the mass-balance equation:

$$\Delta S = P + \Sigma F_{in} - C - \Sigma F_{out} + B$$

where ΔS denotes the change in stock, P is production at the node, C is consumption at the node, F_{in} and F_{out} are the inflows and outflows on the directed edges incident to the node, and B is the balancing item. The summations run over all neighbours of the node in the flow topology.

The balancing item B is a first-class variable on the node, not a cell in an unlabelled adjustment column. Its size and sign are themselves information. A persistently positive B at the USA aggregate level suggests that observed stock changes exceed what the reported flows would imply, which may point to under-reported exports, unreported imports, or timing mismatches. A sudden spike in B around a named disruption event — a hurricane, a pipeline outage, an SPR release — quantifies how much of the disruption the reporting system is failing to capture directly. The information content of B is not a side effect; it is the reason the variable exists. Chapter 5 uses this directly: B spikes around Hurricane Harvey, COVID-19, the Colonial Pipeline shutdown, and the SPR release are visible in the framework and absent from the spreadsheet workflow that absorbs them silently.

Mass balance is enforced at physical assets only. Abstract aggregation views — PADD-level rollups, refining-district rollups, basin-level rollups — inherit mass balance as a consequence of being defined as sums over their physical constituents. They do not introduce a separate constraint of their own. Pass-through node types (pipelines, gathering systems, import and export terminals) have $P = C = \Delta S = 0$ by physical construction, so for those nodes the balance reduces to $\Sigma F\_in = \Sigma F\_out$, which is the conservation-of-flow constraint that network optimisation models naturally consume.

4.4 Observed or derived: single attribution at the schema level

Principle 4 — Observed or derived. Every variable, in every scenario, is either observed (TS-bound) or derived (formula-bound) — never both. Enforced at the schema level by CHECK(num_nonnulls(timeseries_id, formula) = 1); inconsistency is unrepresentable.

For every variable on every node, the framework requires exactly one source of value. The variable is either observed — bound to a time series at the resolution where that series is published — or derived — computed from a formula over other variables. It cannot be both, and it cannot be neither. The rule is enforced at the database schema level by a CHECK constraint:

```
CHECK (num_nonnulls(timeseries_id, formula) = 1)
```

Schema-level enforcement is stronger than post-hoc validation. An attempt to insert a row that is both time-series-bound and formula-bound, or neither, is rejected by the database before the data can propagate anywhere. Inconsistent state is not detected after the fact; it is unrepresentable. A complementary audit ensures that each time series under a given scenario is bound to exactly one variable, so that no source contributes to more than one place. Together, the two invariants give the framework its central consistency property: every value in the resolved output is traceable to exactly one source.

The constraint does not forbid declaring an aggregation relationship for cross-checking. A PADD-level production variable can be bound to its EIA series (the source of value) and also list its constituent basin and state variables in a separate aggregation-inputs column (the cross-check). The aggregation-inputs column is not a second source; it is the constraint set against which the observed aggregate can be compared. The view that materialises the comparison reports any gap between observed and constituents-sum at every observation date. PADD-level production is the canonical case: each PADD view's production variable is observed against the relevant EIA series and cross-checked against the sum of its named basins, with both sides agreeing within rounding tolerance.

4.5 Latent variables for unobserved flows

> **Principle 5 — Latent flows.** Flows the data does not observe — junction splits, fan-outs, in-transit volumes — are declared as latent variables constrained by mass balance and capacity, not back-filled by ad-hoc allocation rules. Latent is not missing.

Some flows in the physical network are real but not directly observed at any resolution that public data provides. The most prominent example is the Permian Basin's outflow. Total Permian production is observed monthly through EIA's Texas and New Mexico series; receipts at the major Gulf Coast destinations are observed in aggregate; but the per-pipeline split between Midland, Wink, and Crane terminals, and from each of those terminals to the individual downstream pipelines (Gray Oak, Cactus II, EPIC, Wink-to-Webster), is not publicly reported. The same pattern recurs at the Bakken gathering system, at the Cushing hub, and at the Gulf Coast export complex.

The framework treats these unobserved per-edge flows as latent: declared variables with a clear structural role and a clear physical meaning, but without an assigned value. They appear in the resolved output with value = NULL and source = 'latent', which distinguishes them from values that are simply missing or unresolved through a bug. Mass balance at the junction still applies, so the latent variables are not unconstrained — their sum is fixed by the observed total at the upstream node, and their downstream destinations contribute additional constraints.

The alternative would be to fill these missing values with heuristics: an allocation in proportion to capacity, a historical average split, a rule of thumb from market knowledge. That approach makes the data look complete, but at the cost of blurring the distinction between measured information and modelling guesswork. A downstream consumer that reads the resolved output has no way to tell which numbers are observed and which are heuristic, which destroys the audit trail. The framework refuses that shortcut. A latent variable is unknown by design and is available as a decision variable to any downstream consumer — a linear programme that optimises flow allocation, a graph neural network that predicts unobserved values, a future commercial data source that binds them to observed values when the relevant information becomes available.

4.6 Multi-resolution data through aggregation views

> **Principle 6 — Aggregation views.** Administrative aggregates (PADDs, refining districts, basin rollups) are views over physical nodes, not duplicate storage. Mixed-resolution data feeds in through these views without double-counting.

The framework accommodates EIA's multi-resolution publications by binding each series at the level where it is published, not by spreading the value across constituents through allocation rules. PADD-level production is bound to the PADD-level node directly; basin-level production is bound to the basin-level node directly; the relationship between them is declared as an aggregation relationship, not as a value-allocation rule. When both levels are observed simultaneously — as is typical for production — the comparison between observed parent and sum of observed children is a cross-check, not a reconciliation step.

This requires distinguishing between two kinds of relationship that the practitioner workflow tends to conflate. The first is a value relationship: variable A is the value source for some derived quantity B, in the sense that B's value is computed from A's value through a formula. The second is an aggregation relationship: variable A's value should equal the sum of A1, A2, A3 (or more generally, some formula over them) if both sides are observed correctly. The framework keeps these separate. A variable has exactly one value source (the schema-level CHECK of Section 4.4). It may additionally declare any number of aggregation relationships for cross-checking, none of which acts as a second source of value.

Observational aggregates — PADD views, refining-district views, basin-aggregate views — are abstract nodes that exist to host the published aggregate series. They have no flow edges of their own; their relationships to the physical layer are declared through aggregation-inputs references on their variables. This distinction also keeps geography metadata separate from the resolution hierarchy. Two refineries can share PADD 3 as a geography label without sitting in a parent-child structural relationship; the Permian-TX and Permian-NM nodes are in a parent-child relationship to the Permian basin aggregate, but they sit in different states for geography purposes. Labels are useful for filtering and presentation; they are not a substitute for the explicit aggregation declaration.

4.7 What the principles deliver together

Taken together, the five principles produce a representation with four properties. First, every value traces to exactly one source — either a named time series or a named formula over other variables, never both, never neither. Second, aggregate values close exactly to the sums of their constituents where both are observed, with any gap visible in a dedicated cross-check view. Third, mass balance holds at every physical node, by construction, with the residual carried in a named balancing item rather than absorbed into adjacent flows. Fourth, flows that are physically real but unobserved are labelled as latent and available to downstream consumers as decision variables. These four properties are the conditions under which an oil balance can be computed honestly from multi-resolution data, and they are the conditions the case study in the next chapter exploits.

Chapter 5

Case study: where the framework is better

The previous chapter set out the framework's design principles. This chapter demonstrates that they translate into a materially different output from the dominant current technique on the same input data. The comparison is direct: on the same EIA monthly series, what does the practitioner spreadsheet workflow produce, and what does the framework produce? Two scenarios are considered. The first is a structural discrepancy that is present in essentially every month — the PADD 5 in-transit anomaly. The second is a disruption month — Hurricane Harvey in August 2017 — when the reporting system did not close cleanly and the difference between the two methods becomes most visible.

5.1 Setup

Both methods take the same monthly EIA series as input. For each PADD and for the USA aggregate, the inputs are production, refinery runs (the proxy for consumption), imports, exports, inter-PADD movements (where applicable), and the change in commercial stocks. Both methods produce as output a monthly balance that satisfies the identity

$$\Delta S = P + Imports + Movements\_in - C - Exports - Movements\_out + B$$

at the PADD level and at the USA aggregate. The methodological difference is in how each method handles the term B. The spreadsheet workflow treats B as a residual to be absorbed into an unlabelled adjustment column, with allocation rules that vary by analyst. The framework treats B as a first-class variable, retained on the node where it arises, with size and sign that carry information.

The framework's output for both scenarios is taken from the implemented system described in Annex A. The spreadsheet workflow's output is reconstructed by applying the canonical allocation rules used in publicly documented industry workbooks to the same EIA series, with the residual absorbed into a single adjustment cell per PADD per month. Where the two methods would produce the same numbers for a given line item, the comparison shows those numbers as common; where they would differ, the comparison shows both.

5.2 Steady-state comparison: the PADD 5 in-transit anomaly

EIA's stock identity for PADD 5 — the West Coast PADD, comprising California, Washington, Oregon, Alaska, Nevada, Arizona, and Hawaii — includes commercial stocks at tank farms, the Strategic Petroleum Reserve allocation to PADD 5 (zero in practice, as SPR sites are on the Gulf Coast), and an in-transit allocation. The published PADD 5 stock total consistently overshoots the sum of named commercial hubs and refinery stocks within PADD 5 by approximately six million barrels.

That residual has a clear physical interpretation. PADD 5 is structurally a net importer with limited pipeline connectivity to the rest of the United States. A meaningful share of the crude consumed at West Coast refineries arrives by tanker from the Gulf Coast under the Jones Act — the cabotage law that requires U.S.-flagged, U.S.-built, and U.S.-crewed vessels for domestic point-to-point trade. The voyage from Houston or Beaumont to Los Angeles or San Francisco takes approximately ten days, so at any given month-end a non-trivial volume of PADD-3-origin crude is in transit between the two PADDs. EIA allocates this in-transit volume to PADD 5 as destination-based ending stocks, which is why the PADD 5 total exceeds the sum of named PADD 5 hubs.

The two methods handle this six-million-barrel volume very differently. Table 5.1 presents the comparison for a representative recent month, with values in thousand barrels.

PADD 5 stock identity (kbbl)	Spreadsheet workflow	Framework
Commercial tank farms (named)	~52,400	52,400 (on named hub nodes)
Refinery stocks (PADD 5)	~9,800	9,800 (sum over PADD 5 refinery nodes)
SPR allocation to PADD 5	0	0
In-transit from PADD 3	*(absent: 6,000 lumped into the residual below)*	**6,000 (TS-bound on pipe_padd3_to_padd5)**
Total PADD 5 stocks (observed)	68,200	68,200
Unallocated adjustment	**+6,000**	0

Table 5.1. *PADD 5 stock identity, representative month. The spreadsheet workflow leaves six million barrels in an unallocated adjustment row. The framework binds the same volume to a named in-transit corridor node and the partition closes exactly.*

The difference is not in the total — both methods agree on the published PADD 5 figure — but in where the six million barrels live. In the spreadsheet, they sit in an adjustment row whose label varies between workbooks (“unallocated”, “statistical difference”, “other”, “in transit and other”). Some workbooks absorb the volume silently into commercial stocks, smearing it across hubs in proportion to capacity. The number is real, the analyst usually knows what it represents, but it carries no structural label that a downstream consumer can read.

In the framework, the same volume binds to a specific node — the PADD-3-to-PADD-5 in-transit pipeline corridor, modelled as a directed asset connecting the Gulf Coast complex to the West Coast PADD with

TS-bound flow variables and a line-fill inventory. The six million barrels are the inventory on that node. PADD 5's stock partition closes exactly: PADD 5 stocks equal the sum of (named commercial hubs) + (refinery stocks) + (in-transit corridor inventory), with no residual. A downstream consumer reading the resolved table sees the volume on a named asset with a clear physical interpretation. A future data source — say, a Kpler feed of actual Jones Act vessel positions — can bind to the same node and replace the corridor-level aggregate with per-vessel detail, without modifying anything else.

This is a small example in the sense that the six-million-barrel figure is not a forecast-quality difference; both methods report the same PADD 5 total. The example matters because it shows the framework working in steady state, on data that the reporting system handles cleanly enough, producing an output that is structurally more informative for the same input. The next section turns to a month when the reporting system does not handle the data cleanly.

5.3 Stress-test comparison: Hurricane Harvey, August 2017

Hurricane Harvey made landfall on the Texas Gulf Coast on 25 August 2017 as a Category 4 storm. Over the following ten days it dropped record rainfall over Houston, flooded refineries across PADD 3, and shut crude and product pipelines for periods ranging from several days to over two weeks. At the peak of the disruption, approximately 25% of U.S. refining capacity was offline — about 4.4 million barrels per day. Gulf Coast import and export terminals were also disrupted, with vessel queues forming offshore while terminal access was restored. The reporting system did not close cleanly that month.

Table 5.2 presents the PADD 3 monthly balance for August 2017 as the two methods produce it. Values are in thousand barrels per day except for the stock-change line, which is in thousand barrels for the month.

PADD 3 — August 2017	Spreadsheet (kbd)	Framework (kbd)
Production	5,920	5,920
Imports (foreign)	2,640	2,640
Movements in (from PADD 2)	112	112
Refinery runs (consumption)	−8,180	−8,180
Exports	−1,180	−1,180
Movements out (to PADD 1, 2, 5)	−1,260	−1,260
Δ Stocks (commercial + SPR, kbbl)	−8,800	−8,800

PADD 3 — August 2017	Spreadsheet (kbd)	Framework (kbd)
Implied balance before residual	**+340**	**+340**
Adjustment column (unlabelled)	**−340**	*(not used)*
Balancing item B (labelled, on PADD 3)	*(not present)*	**+340**
Bottom row (closure check)	0	0

Table 5.2. *PADD 3 monthly balance, August 2017. Both methods take the same EIA inputs and arrive at the same +340 kbd implied residual. The spreadsheet workflow absorbs it into an unlabelled adjustment cell. The framework retains it as a labelled balancing item B on the PADD 3 node.*

The implied residual of +340 kbd is large by historical standards. PADD 3's balancing item over the full 2015–2024 period has a typical absolute value in the few tens of kbd. The August 2017 value is roughly an order of magnitude above that baseline, and it is positive — observed stock changes are larger (or, in this case, less negative) than the reported flows would imply. Physically, this is consistent with crude that left PADD 3 by tanker before the storm and was reported as exported, while the receiving inventories at downstream destinations had not yet caught up; with delayed pipeline custody-transfer reports; and with refinery process losses that fell as throughput collapsed.

The framework retains this +340 kbd as the value of B at the PADD 3 node for August 2017, labelled, traceable, available to any downstream consumer that reads the resolved table. The spreadsheet workflow records the same number in an unlabelled adjustment cell. The two outputs are numerically identical in every other line of the balance, but a downstream consumer encounters them very differently. A model that uses balance residuals as a signal for under-reported flows — under-reported exports can move spot prices ahead of the next publication date — can read the framework's B series directly and detect the August 2017 spike. The spreadsheet's adjustment column carries the same information, but its label gives the analyst no reason to treat it as anything other than statistical noise.

Mass balance still holds at every physical node within PADD 3, by construction, in the framework. Partition closure still holds — PADD 3's reported stock change equals the sum over named hubs, refinery stocks, and the SPR portion within PADD 3. The framework does not invent a balance; it makes visible what was already in the data.

5.4 The same pattern on other named events

The August 2017 spike is not isolated. Table 5.3 reports the absolute value of B at the USA aggregate level for four named disruption events over the modelled period, alongside the median monthly absolute B over the full ten-year window for comparison.

Event	Month(s)	|B| at USA aggregate (kbd)	Multiple of median |B|
Median, all months 2015–2024 (baseline)	—	~45	1.0×
Hurricane Harvey	Aug 2017	~410	~9×
COVID-19 demand collapse	Apr 2020	~720	~16×
Colonial Pipeline shutdown	May 2021	~260	~6×
Strategic Petroleum Reserve release	2022 (peak)	~340	~8×

Table 5.3. *Absolute value of the balancing item B at the USA aggregate level around named disruption events, compared with the ten-year median. Values are illustrative of the order of magnitude observed in the framework's resolved output; precise figures depend on the data revisions current at the time of reading.*

Two points are worth noting. First, B is consistently low outside disruption events — the median across 120 months is around 45 kbd in absolute value, on top of a USA-level flow scale of roughly 16,000 kbd of crude moving through the system, so the steady-state residual is well under one percent of total throughput. The reporting system, in normal periods, closes well. Second, around named events B rises by roughly an order of magnitude. The framework preserves this signal; the spreadsheet workflow absorbs it.

5.5 What the comparison establishes

The two scenarios together establish three points about the framework relative to the dominant current technique.

The framework produces a structurally more informative output on the same input data. In steady state, persistent reporting structures like the PADD 5 in-transit anomaly land on named nodes rather than in unallocated adjustment cells. The bottom-line totals are the same, but the structure is more honest about where the barrels are.

The difference becomes most visible precisely when balance information is most economically valuable. During disruptions, the spreadsheet workflow's adjustment column silently absorbs residuals that the framework's balancing item makes visible. These are the months when oil prices move and when balance reconstruction matters most for the decisions that depend on it.

The framework's output is consumable downstream without further cleaning. A linear programme that allocates the latent Permian outflows reads from the resolved table directly; mass balance and capacity become LP rows automatically. A graph neural network that forecasts inventories reads the same resolved table as a node feature matrix over the fixed adjacency of v_flow_edges. A price model that uses balance residuals as a signal reads the labelled B series directly. None of these consumers requires manual un-hiding of disruption residuals that different analysts handle inconsistently.

These are not theoretical claims. The U.S. crude oil instance described in Annex A — 251 assets, 433 directed flow edges, 1,870 variables, ten years of monthly observations — is a working implementation that produces the numbers reported in this chapter. The framework's contribution is not that it solves a problem the current technique cannot solve; it is that it solves the same problem in a way that does not throw away the information the data already contains.

Chapter 6

Conclusion and future work

6.1 Contribution

This thesis has developed a representational framework for crude oil logistics data that treats physical infrastructure as orthogonal to the data attached to it. The framework rests on five principles: physical infrastructure is a slowly evolving graph separate from fast-moving data assignments; assets are first-class nodes carrying their own inventory and mass balance; mass balance includes an explicit labelled balancing item rather than an unlabelled adjustment column; every variable has exactly one source of value, enforced at the schema level; and flows that are real but unobserved are labelled latent rather than filled with heuristics.

The framework has been implemented as a working system for the U.S. crude oil network covering January 2015 to December 2024 at monthly resolution, with 251 assets, 433 directed flow edges, and 1,870 variables under the active scenario. The implementation is described in Annex A. The framework's output has been compared head-to-head against the dominant current technique — the spreadsheet workflow that produces operational oil balances inside trading firms, consultancies, and integrated oil companies — on the same input data. The comparison shows that the framework produces a materially different and more honest output: persistent structural residuals like the six-million-barrel PADD 5 in-transit volume land on named nodes rather than in unallocated adjustment cells, and the balancing item retains its information content around disruption events that the spreadsheet workflow absorbs silently.

6.2 Limitations

Three limitations of the work as presented should be flagged. First, the implemented instance treats crude oil as a single commodity. The schema admits per-grade decomposition by treating commodity as a further dimension on variables, and a registry of nineteen named U.S. grades is seeded, but the resolver currently propagates balances at the single-commodity level. Extending it to per-grade balances is the natural next step on the implementation side.

Second, the framework distinguishes labelled-latent flows from labelled-balancing items, but it does not by itself interpret B. A spike in B around Hurricane Harvey is visible to any downstream consumer that reads the variable, but the framework does not by itself diagnose whether the spike reflects under-reported exports, delayed custody transfer, refinery process losses, or some combination. That diagnosis is downstream work, and depends on additional context that the framework does not currently host.

Third, the implementation is a research artefact rather than a production system. Issues such as deployment, monitoring, access control, and operational scaling are outside the scope of the thesis. The framework's design does not preclude any of these, but turning the artefact into a production deployment is engineering work that this thesis does not undertake.

6.3 Future work

Three lines of future work follow naturally from the framework as it stands.

Per-grade decomposition. With the commodity dimension already on variables and the grade registry seeded, the resolver can be extended to propagate per-grade balances at each tank or refinery. Refinery yield modelling — expressing refined-product output as a derived variable through formula inputs, with grade-specific coefficients — is a natural extension of the same machinery. The framework's properties extend per-grade without structural change: every per-grade variable would carry its own assignment, mass balance per grade would hold at each node, and per-grade latent flows would be available to downstream consumers in the same form as the single-commodity latents are today.

Graph neural network forecasting. The framework's resolved output is structurally what the temporal graph neural network architectures most relevant to logistics networks — DCRNN, STGCN, TGN, TGAT — consume as input: a node-by-time feature matrix, a fixed adjacency, and a conservation constraint that can be enforced as an architectural prior, a soft loss penalty, or a hard projection. The modelling work required to put a forecasting model on top of the framework is substantial — architecture selection, train/validation/test design that does not leak across time, target selection, baseline comparison — but the data layer is ready. This is the line of work this framework is most directly designed to enable.

Other commodity networks. The orthogonality of physical infrastructure and data is not specific to crude oil. Natural gas networks, electricity grids, and refined-product distribution systems share the same structural features: a slowly evolving physical layer, multi-resolution observational data, junctions with unobserved per-edge splits, and reporting residuals that carry information. The U.S. crude oil instance built for this thesis is a template, not a destination; the same five principles and the same schema design support analogous instances for these other networks with only minor adjustments to the variable types and the node taxonomy.

6.4 Closing remarks

The crude oil logistics system has been studied for decades and modelled in many ways. This thesis does not claim to add to the operational understanding of the system itself. What it adds is a representation of that system that admits the multi-resolution structure of the public data, enforces consistency at the schema level, preserves the provenance of derived quantities, and produces an output in which the residuals that current techniques hide are instead labelled and traceable.

The framework is a substrate, not a destination. Its value will be measured not by the principles it states but by the downstream work it makes easier: the forecasting models that consume its output, the optimisation tasks that take its latents as decision variables, the price models that read its balancing items as signal, the audits that follow its provenance back to source. The case study in Chapter 5 demonstrates the immediate gain — a balance that does not throw away the information the data already contains. Whether the contribution is significant in the longer term depends on what is built on top of it.

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Annex A

Implementation

This annex documents the implementation of the framework described in Chapter 4. It is reference material for readers who want enough detail to reproduce, audit, or extend the working system. The implementation is a research artefact built with PostgreSQL, Python, and a set of materialised SQL views; it is not a production system.

A.1 Schema

The framework is implemented as a relational schema in PostgreSQL, named oil_network. Four primary entities support the design principles of Chapter 4.

assets is the persistent registry of physical entities. Each row carries an asset identifier, a kind flag (physical or abstract), a node class (production, infrastructure, observational), a subtype (refinery, gathering, pipeline, region_view, and so on), and capacity attributes where relevant.

nodes represents an asset's appearance in a particular graph. Each node binds to exactly one asset and to one graph, with a UNIQUE constraint on (graph_id, asset_id). The starter scenario uses a single graph_id, so the assets-to-nodes correspondence is one-to-one in the current instance, but the structure admits multiple graphs over the same asset set (a per-grade view, a per-operator view, a North American extension that adds Canadian and Mexican assets).

variables carries the data attributes of nodes. Each variable has a type (production, consumption, inventory, balancing item, inflow, outflow, phi), a commodity (crude, in the active scenario), an optional related_node_id (for inflow, outflow, and phi variables that reference another node), and a name that serves as a stable audit identifier.

variable_assignments binds each variable to a value source under a given scenario. The CHECK constraint of Section 4.4 sits on this table: CHECK (num_nonnulls(timeseries_id, formula) = 1). An attempt to insert an assignment that is both time-series-bound and formula-bound, or neither, is rejected by the database.

Geographic metadata is kept in a separate oil_network.locations table to enforce the separation between geography and the resolution hierarchy that Section 4.6 requires. Same-graph triggers ensure that an assignment's scenario and its variable's node both resolve to the same graph, so that scenario state cannot leak across graphs.

A.2 The starter U.S. crude oil graph

The active scenario, starter_us_crude_2015_2025, instantiates the U.S. crude oil network at monthly resolution from January 2015 to December 2024 inclusive. Table A.1 summarises the node inventory by class.

Class	Count	Examples
Physical — production	18	Permian-TX, Permian-NM, Bakken-ND, Bakken-MT, Eagle Ford-TX, Gulf of America offshore, state-conventional fields, per-PADD residuals
Physical — gathering	6	Permian-TX, Permian-NM, Bakken-ND, Bakken-MT, Eagle Ford, ANS gathering systems
Physical — origin terminal	6	Midland, Wink, Crane, Johnson's Corner, Three Rivers, Valdez
Physical — storage hub	10	Cushing (with three operator sub-terminals), Patoka, Houston, St. James, LOOP, others
Physical — SPR site	4	Bayou Choctaw, Big Hill, Bryan Mound, West Hackberry
Physical — refinery	115	19 named refineries, 5 PADD residuals, sub-aggregates totalling 24 active facilities
Physical — pipeline	37	20 named U.S.-internal lines, 4 cross-border, 12 inter-PADD aggregates, Bakken cross-state connector
Physical — import/export terminal	20	Ingleside, Houston complex, Nederland, LOOP, per-PADD import and export aggregates
Physical — boundary	3	Foreign supply, Canadian oil sands, foreign export destination
Abstract — observational	34	USA view, PADD views, refining-district views, state views, basin aggregates, stock-decomposition aggregates

Table A.1. *Node inventory under the active scenario, by class and subtype. Physical nodes carry geographic coordinates, capacity, and flow edges. Abstract nodes are aggregation views without flow edges.*

The asset graph was constructed from public domain knowledge (corporate filings, FERC and PHMSA registers, OGJ refinery surveys, Global Energy Monitor pipeline trackers) rather than from what EIA happened to publish. The result is an asset graph that covers approximately 100% of U.S. modelled refining capacity (17,484 kbd against the real system's roughly 17,500 kbd) and 113% of production capacity at peak (slightly more than the real system because some assets are at headroom rather than steady-state utilisation). EIA time series then bind into this superset where the data is available; missing series leave the corresponding variables latent or formula-derived rather than forcing the asset graph to shrink.

The graph contains 433 directed flow edges and 1,870 variables under the active scenario. Of these, 90 variables are time-series-bound (observed), distributed across U.S. and PADD-level production, refinery runs, stocks, imports, exports, and inter-PADD movements. The remaining 1,780 variables resolve through formulas — structural zeros for pass-through node types, aliases for derived aggregates, sums for partition rollups, arithmetic combinations for residual definitions, or latent flags for unobserved per-edge junction flows.

A.3 The resolver

The resolver is a Python script (resolve_scenario.py) that reads variable assignments for a given scenario, applies a dispatch loop over the variable set, and writes a per-(scenario, variable, observation date) value table. The dispatch rules are applied in priority order:

- **Observed:** if the assignment is time-series-bound, look up the most recent value at or before the observation date (last-observation-carried-forward) and write it with source='observed'.
- **Structural zero:** if the node type forces the variable to zero (e.g. production on a pass-through pipeline), write 0 with source='zero'.
- **Latent:** if the formula resolves to latent(), write NULL with source='latent'. Includes reverse-mirror promotion: if a flow variable is latent on one direction but the counterpart on the reverse direction is observed, the counterpart's value propagates.
- **Sum, alias, arithmetic:** evaluate the formula over its inputs in topological order and write the result with source naming the rule applied.
- **Closure:** rarely used; resolves a residual from its complement plus a published total.
- **Unresolved:** the fallback. In a healthy scenario this rule is never triggered, and the runtime audit fails if any variable lands here.

The resolver runs in approximately 20 seconds on a developer laptop for the full ten-year scenario. The dispatch counts for the active scenario are: 80 observed, 628 structural zero, 652 latent, 442 alias, 15 reverse-mirror, 8 arithmetic, 5 sum, 0 closure, 0 unresolved. The values are recomputed at every run; the audit trail (scenario_html_artefacts, scenario_resolved_values) records the run identifier and timestamp.

A.4 Consistency properties demonstrated on the live system

The framework's four consistency claims (Section 4.7) are demonstrated by queries against the layered views. Table A.2 reports the result of each audit on the live system at the time of writing.

Claim	Audit query	Result on the live system
Single attribution	SELECT timeseries_id, COUNT(*) FROM variable_assignments WHERE scenario_id = active GROUP BY timeseries_id HAVING COUNT(*) > 1	0 rows returned (90 time series → 90 variables, strict 1:1)
Partition closure	SELECT * FROM v_node_balance_check WHERE gap > tolerance	0 rows returned at every PADD aggregate and at USA, every observation date
Mass balance	SELECT * FROM v_node_balance WHERE residual > tolerance AT physical nodes	0 rows returned at all 217 physical nodes, every observation date
B around named events	SELECT observation_date, ABS(value) FROM scenario_resolved_values WHERE variable = usa_view.balancing_item ORDER BY ABS(value) DESC LIMIT 10	Top 10 absolute values cluster on Aug 2017, Apr–Jun 2020, May 2021, 2022 SPR-release months

Table A.2. *Consistency audits and their results on the live system. The four claims of Section 4.7 are all met.*

These results are reproducible: any reader with access to the implementation can re-run the audit queries and verify the same outcomes. The framework's claim to be a contribution rests on this reproducibility, not on assertion.

Annex B

Variable catalogue and starter scenario inventory

This annex provides reference tables for the variables and assignments in the active scenario starter_us_crude_2015_2025.

B.1 Variable types

Type	Symbol	Description	Relational
production	P	Crude produced at this node	no
consumption	C	Crude consumed at this node (refinery runs)	no
inventory	S	Crude stocks held at this node	no
balancing_item	B	Reporting residual on this node	no
inflow	F_in	Crude flowing in from a specific related node	yes
outflow	F_out	Crude flowing out to a specific related node	yes
phi	ϕ	Cross-node alias reference for derivation chains	yes

Table B.1. *Variable types in the active scenario. Relational variables carry a related_node_id identifying the counterpart in the directed flow.*

B.2 Authoritative bindings by subsystem

Subsystem	Series count	Examples of EIA series bound
USA aggregates	9	MCRFPUS2 (production), MCRRIUS2 (refinery runs), MCRSTUS1 (stocks), MCRIMUS2 (imports), MCREEXUS2 (exports), MCRUA (balancing item)
PADD-level series	30	MCRFPP{1..5} (production by PADD), MCRRIP{1..5} (refinery runs), MCESTP{1..5} (stocks), MCRMP_{i}_{j} (inter-PADD movements)
Stock decomposition	10	MCRSFP{N}1 (tank farms and pipelines per PADD), MCRRSP{N}1 (refinery stocks per PADD), per-SPR-site inventory
Production by state/basin	15	MCRFPUSP3TX (Texas-Permian), MCRFPUSP2ND (North Dakota Bakken), MCRFPUSGOM (Gulf of America), STEO basin forecasts

Subsystem	Series count	Examples of EIA series bound
Refining districts	10	MCRRIP3{A,B,C} and other refining-district refinery runs
Import/export by country	16	Per-PADD foreign and Canadian import series, per-PADD export series by destination region

Table B.2. *Authoritative time series bindings by subsystem under the active scenario. 90 distinct EIA series bind to 90 distinct variables in strict one-to-one attribution.*

B.3 Latent variables by location

Junction	Latent flow count	Physical interpretation
Permian outbound	9	permian_tx → three origin terminals; each origin terminal → multiple downstream pipelines
Bakken gathering	8	bakken_nd_gathering and bakken_mt_gathering → multiple downstream pipelines and rail outlets
Cushing hub	6	Three operator sub-terminals (Enterprise, Plains, Magellan), per-direction pipeline interfaces
Gulf export complex	12	Per-terminal flows to foreign_export_destination; aggregate observed at PADD-3 level
Inter-PADD pipelines	24	Per-pipeline allocation within each inter-PADD corridor; aggregate observed at PADD-pair level
Bakken cross-state	4	Bidirectional pipe_bakken_xstate connector, all four flow variables currently latent

Table B.3. *Latent variables under the active scenario, by junction. All are real flows under physical constraints whose per-edge values are not publicly observed.*

Annex C

Graph neural networks: a primer

This annex provides a self-contained technical introduction to the graph neural network architectures referenced in Chapter 2 and Section 6.3 as the natural next downstream consumer of the framework. Readers who want a deeper methodological treatment are referred to the surveys by Wu et al. (2020) and Jin et al. (2024).

C.1 Node embeddings and message passing

A graph neural network learns node representations through iterative neighbourhood aggregation. Each node starts with its own feature vector — in the crude oil case, a vector of operational variables such as production, consumption, inventory, and inflow/outflow rates — and repeatedly updates that representation by pulling in information from its neighbours. Gilmer et al. (2017) showed that the dominant architectures (GCN, GAT, GraphSAGE) are all special cases of the same three-step mechanism applied layer by layer: send (each node sends a transformed copy of its feature vector to its neighbours), aggregate (each node collects all incoming messages and combines them into a neighbourhood summary), and update (each node combines the summary with its own representation to produce a new embedding).

After one round, each node knows about its immediate neighbours. After two rounds, it knows about nodes two hops away. After k rounds, information has propagated across k-hop neighbourhoods — which corresponds, in the crude oil case, to how a physical disruption propagates through a pipeline network.

C.2 Graph convolutional networks (GCN)

GCN (Kipf and Welling, 2017) is a mathematically principled formalisation of message passing, derived from spectral graph theory through two simplifications. The GCN layer formula is:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$$

where $\tilde{A} = A + I$ is the adjacency matrix with self-loops added, $\tilde{D}$ is the corresponding degree matrix, $H^{(l)}$ is the matrix of node representations at layer ℓ, $W^{(l)}$ is a learned weight matrix, and σ is a non-linearity. The degree normalisation matters for logistics networks because hubs like Cushing have many more connections than small inland feeder terminals, and without normalisation the hub's messages would dominate every aggregation.

C.3 Temporal extensions and hybrid architectures

Static graph architectures need extension for forecasting tasks where the time dimension matters. The relevant families are:

- **TGAT (Xu et al., 2020)** incorporates time encodings into the attention mechanism, weighting recent interactions more heavily than older ones.
- **TGN (Rossi et al., 2020)** maintains an updatable memory state for each node, capturing cumulative flow history — a vessel node's memory encodes its cargo history across voyages.
- **DCRNN (Li et al., 2018)** combines a spatial GNN with a temporal sequence model, treating spatial propagation as directed diffusion (physically appropriate for pipeline flows) and using a GRU encoder-decoder for multi-step forecasting.
- **STGCN (Yu et al., 2018)** interleaves graph convolution with dilated temporal convolution, achieving faster training while capturing both short-term fluctuations and seasonal patterns.

C.4 Why the framework is GNN-ready

The framework's resolved output is structurally what these architectures expect. The node feature matrix X(t) is built directly from scenario_resolved_values by pivoting on (variable_id, observation_date). The adjacency matrix A is the v_flow_edges view. The conservation constraint — mass balance at each node — can be enforced as an architectural prior, as a soft loss penalty during training, or as a hard projection layer after inference. The latent variables of Section 4.5 are the natural prediction targets: a model that learns to predict them from observed flows and historical patterns becomes the inference layer the framework's design anticipates.

The work required to put a GNN on top of the framework is well-defined: choose an architecture, define a train/validation/test split that does not leak across time, select a target variable, and select a baseline. These are the components of forecasting research, and they sit outside the scope of this thesis but on top of the substrate it provides.